

AD7190

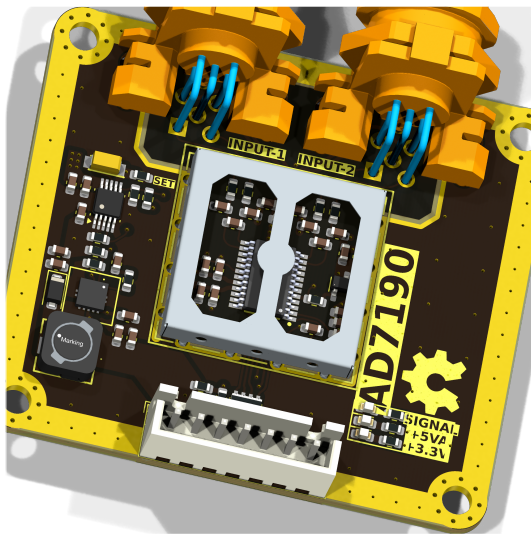
Variant:

2026-10-09

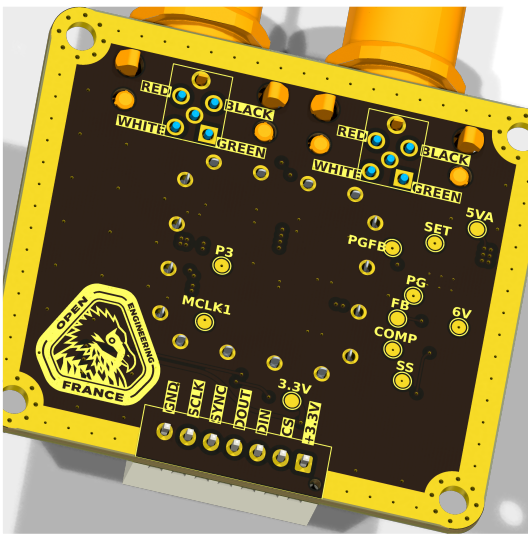
Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11		21		31	
2	Block Diagram	12		22		32	
3	Project Architecture	13		23		33	
4	AD7190_Weight_Sensor	14		24		34	
5	LT3042_6V_to_5VA	15		25		35	
6	TPS61086_3.3V_to_6V	16		26		36	
7	Holes, Fiducials	17		27		37	
8	Connectors	18		28		38	
9	Power - Sequencing	19		29		39	
10	Revision History	20		30		40	

TOP VIEW



BOTTOM VIEW



NOTES

OPEN_WEIGHT - AD7190 24-bit load-cell front-end, part of OPEN THRUST LAB.
Single +3.3 V host supply, SPI interface, 2 x 4-wire bridge inputs.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 09-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

+3.3V host -> TPS61086
6 V -> LT3042 +5VA (AVDD + excitation)

DEBUG NOTE:

TP on every rail.
Wait >= 600 ms (PG) before ADC init.

DESIGN NOTE:

SPI mode 3,
SCLK <= 5 MHz,
RDY on DOUT.

DESIGN NOTE:

REFIN1 = bridge excitation: ratiometric.

LAYOUT NOTE:

AFE under the WE-SHC shield H5, boost SW node away.

	Comments:	Company: OPEN THRUST LAB	Variant:	Git Hash: b3867f9
	Title: OPEN_WEIGHT_AD7190	Board Name: AD7190	Project Name: OPEN_WEIGHT	
	File Name: OPEN_WEIGHT.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 10